

DIGEST Analysis Workflow Documentation

1. Introduction

This document provides a detailed account of the workflow for the DIGEST trial interim analysis. It is intended to ensure reproducibility, transparency, and ease of navigation for future analysts or reviewers. The DIGEST trial investigates whether a remote Habitual Remission Programme, combining a Total Diet Replacement (TDR) intervention with digital therapeutic support, can achieve type 2 diabetes remission and clinically significant weight loss in a cost- and resource-effective manner.

The code package and supporting files enable replication of the full analytical pipeline, from raw data ingestion through to the generation of baseline summaries, missing data reports, and interim outputs for the Statistical Analysis Plan (SAP).

2. Local Repository Structure

The project is provided as an RStudio Project (DIGEST.Rproj) packaged in a zip file. The following directories and files are included:

1. scripts/:

Contains three R scripts, each corresponding to a major stage of the analysis pipeline.

- DIGEST_01_data_cleaning.R – Data ingestion, cleaning, processing, and standardisation.
- DIGEST_02_imputation_and_efficity_analysis.R – Data Analysis including computing the primary and secondary endpoints
- DIGEST_03_safety_and_adherence_reporting.R – Safety Analysis and Adherence reporting

2. data/:

Contains all data files required for analysis.

- data/raw/: Raw source files (CSV format). These files must not be modified.
- data/clean/: Intermediate cleaned datasets generated by Script 1.
- data/master/: Final merged analysis dataset (master_dataset.csv), adverse events dataset (adverse_events_analysis.csv), and medications dataset (medications_analysis.csv).
- data/mice_datasets/: Multiple imputation datasets used in subsequent analyses.

3. documents/:

Contains the SAP, protocol, and supporting documentation for reference.

4. output/:

Contains all derived outputs for review. Subfolders include tables, plots, and reports as

generated in Script 2 and 3.

5. DIGEST.Rproj:

The RStudio Project file. Opening this file ensures that relative paths resolve correctly and that scripts can be run without manual adjustment.

6. README.md:

Provides a concise overview of the repository and directory organisation.

3. Software Requirements

- R version: 4.5.1
- RStudio: Desktop version 2025.05.1 (or later recommended)
- Operating system: Scripts were developed on macOS, but are cross-platform compatible.

Required R Packages are installed automatically if missing at the housekeeping section of each script.

4. Workflow Overview

The analytical pipeline is structured into three sequential scripts. Each script must be run in order.

Script 1: Data Preparation and Cleaning (***DIGEST_01_data_cleaning.R***)

- Ingests raw CSV files across 7 datasets (AE, BMI, Medications, etc.)
- Cleans and standardises individual datasets
- Merges datasets by subject ID and timepoint where needed
- Prepares a combined analysis dataset with baseline and follow-up data, and
- Generates a missing data report across key endpoints (weight, HbA1c, BP, waist)

Script 2: Data Analysis (***DIGEST_02_imputation_and_efficacy_analysis.R***)

- Imputes missing data using Multiple Imputation by Chained Equations (MICE)
- Computes within-subject change scores
- Performs the primary endpoint analysis
- Performs the secondary endpoint analyses
- Generates tables and figures to summarise the findings

Script 3: Safety Reporting (***DIGEST_03_safety_and_adherence_reporting.R***)

- Computed the safety analysis of the trial, and
- Adherence and completion summaries

5. Reproducibility Instructions

1. Unzip the package into a local directory (~/Desktop/DIGEST/).
2. Open the project by double-clicking DIGEST.Rproj.
3. Run the scripts sequentially from 01 to 03. Each script is fully annotated and structured into sections.
4. Verify outputs:
 - Cleaned datasets are created in data/clean/.
 - The master dataset is created in data/master/.
 - Summary tables and reports are generated in output/.
5. Do not modify files in data/raw/ to preserve reproducibility.

6. Method and Outputs

The analysis was conducted in a structured manner, beginning with the cleaning and integration of raw datasets to establish a master dataset. The primary efficacy analyses focused on weight change and diabetes remission, assessed through reductions in HbA1c and achievement of clinically significant weight loss at baseline, three, six, and twelve months. Secondary analyses evaluated changes in systolic and diastolic blood pressure, waist circumference, and BMI across the same timepoints. Safety analyses were undertaken using adverse event data, and adherence was assessed through survey completions and treatment exposure records. Missing data were systematically characterised and multiple imputation datasets were prepared to support analyses. Finally, baseline demographic and clinical characteristics were summarised by treatment group, and formatted tables were generated for reporting in accordance with the Statistical Analysis Plan.

The following outputs were created:

- Cleaned datasets (data/clean/)
- Master dataset (data/master/)
- Missing data report
- Baseline demographic summaries
- Primary and secondary analyses reports
- Imputed datasets are stored in data/mice_datasets/
- Formatted tables and plots

7. Notes

- All data cleaning decisions and variable definitions are documented in-line within the R scripts.
- The pipeline is designed to be modular, allowing extension or modification for the final analysis phase.